

Aarav Daga

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EXPERIENCE

Lockheed Martin

Remote

Platform Engineering Intern

May. 2026 – Aug. 2026

- Shipped 9 merged MRs to Panel, a full-stack Go/React MLOps platform serving 7,500 enterprise users
- Cut API latency 60% across 4 pipelines spanning backend API services and frontend workflows
- Built a full-stack job rerun feature on a new Go endpoint generating create-ready job templates, handling GPU type normalization, command wrapper deduplication, and ConfigMap file restoration; tested with Go and Jest
- Eliminated latency spikes under high pod counts by batching per-pod GPU requests behind a 5-concurrent cap
- Reworked utilization-page loading to separate cold-start from background refresh with per-group caching
- Fixed a Go fail-open regression in job-log handling, preserving response-shape contracts on provider errors

Lockheed Martin via Nittany AI Alliance

State College, PA

Software Engineering Intern

Jan. 2026 – Apr. 2026

- Debugged tensor shape errors in custom CUDA rasterization kernels for a novel sonar Gaussian splatting framework generating synthetic underwater imagery from multi-view video, reaching 35.7 dB PSNR and 0.98 SSIM
- Enabled multi-GPU distributed training on 2 NVIDIA GPUs by resolving 11 bugs across parameter sharding, missing `all_reduce` gradient synchronization, NCCL process-group deadlocks, and cross-rank densification
- Eliminated cross-rank parameter divergence with rank-0 stochastic sampling via `dist.broadcast`
- Configured the HPC training pipeline with PyTorch 2.6, CUDA 12.4, and custom-built CUDA extensions
- Identified per-pixel rasterization as the compute bottleneck by benchmarking single- vs. multi-GPU training

American Society of Mechanical Engineers: Assistive Tech

State College, PA

Lead Software Developer

Sep. 2025 – Mar. 2026

- Led a 5-person team building real-time speech-translation glasses on ESP32, achieving <200ms audio latency
- Built the low-latency audio ingestion, processing, and Bluetooth display path in custom C++/Python firmware
- Integrated microphone, OLED, and Bluetooth modules with custom drivers and debounced I/O

PROJECTS

Distributed Key-Value Store | C++, GTest

Jan. & Sep. 2026

- Built a distributed key-value store in C++20 with an ASIO server, and custom length-prefixed binary protocol.
- Designed a storage-engine interface (ordered map with prefix scans) for future persistence and Raft replication.
- Implemented a concurrency-safe transaction layer with deadlock-free money transfers and locks, covered by 46 tests.

x86_64 Microkernel | C, x86 Assembly

Dec. - Jan. 2025

- Built a 1,200-line microkernel booting on real hardware with long mode and 4-level identity-mapped paging
- Implemented an IDT with 32 exception handlers, including error-code handling and stack-frame management
- Designed syscall infrastructure with dedicated ring-3 gates for secure kernel-userspace communication

GitContext – C-Based Developer Tooling | C, JSONL, SQLite

Nov. – Dec. 2025

- Built a Unix CLI for attaching metadata to Git commits through custom file I/O in the `.git/` directory
- Implemented JSONL note storage, a SQLite TODO manager, and post-commit/merge/pull hook automation
- Added diff-safe merge-driver behavior, repository-scoped daemon mode, and CI/CD with GitHub Actions

EDUCATION

The Pennsylvania State University

University Park, PA

B.S. Computer Science, Minor in Computer Engineering

Expected December 2027

- Coursework: Computer Architecture, Networking, Theory of Computation, Data Structures & Algorithms, Systems Programming, Language Principles, Digital Design

TECHNICAL SKILLS

Languages: C, C++, Go, Python, Java, JavaScript, TypeScript, Bash, x86 Assembly

Systems: Linux, Kubernetes, Docker, TCP/IP, Networking, Multithreading, Concurrency, Memory Management

OS & Architecture: OS internals, computer architecture, file I/O, processes/threads, performance profiling

Embedded: ESP32, Raspberry Pi, Firmware, Device Drivers, SPI/I2C, Bluetooth/BLE, Low-Latency Systems

Tools: Git, gdb, QEMU, NASM, GTest, Jest, CUDA, NCCL, PyTorch, SQLite, Winsock2